

# **Retail Pulse AI-Powered Retail Analytics & Inventory Optimization Platform**

Turn retail transaction data into actionable forecasts, customer segments, churn risk signals, and inventory decisions.

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# 1. Project Overview

## Vision & Objectives

- Build an end-to-end retail analytics platform that converts raw UK Online Retail data into operational intelligence.
- Enable executives and supply chain teams to understand customers, forecast demand, detect churn risk, and optimize inventory.
- Deliver actionable dashboards with real product-level insights instead of synthetic placeholder data.

## Target Users / Use Cases

- Retail operations managers monitoring stock and reorder priorities.
- Marketing teams segmenting customers for targeted campaigns.
- Data analysts validating demand trends and seasonality.
- Business leaders evaluating revenue, churn, and customer health.

## Business Value Delivered

- Reveals hidden customer segments and revenue contributions.
- Supports proactive inventory planning using EOQ and safety stock.
- Provides churn risk classification for retention campaigns.
- Empowers forecasting decisions with historical demand and 30-day outlooks.

## Non-functional Goals

- Accuracy: capture meaningful forecast patterns and segment structure.
- Deployability: streamline into a deployable Streamlit app.
- Usability: interactive visual dashboard with search, filters, and export.
- Scalability: support 1M+ records from combined retail datasets.

## 2. Key Features

| ID   | Feature                | Description                                                    | Acceptance Criteria                                                         |
|------|------------------------|----------------------------------------------------------------|-----------------------------------------------------------------------------|
| F-01 | Data Exploration       | Load and profile combined Online Retail datasets.              | 1,067,370 rows loaded, missing-value summary generated, dataset cleaned.    |
| F-02 | Customer Segmentation  | RFM-based clustering with 6 business segments.                 | 6 clusters created, segment metrics displayed, cluster names interpretable. |
| F-03 | Demand Forecasting     | Historical sales + 30-day forecast with trend and seasonality. | Forecast chart and table available, metrics reported.                       |
| F-04 | Churn Prediction       | XGBoost churn classifier with risk segmentation.               | Churn probability distribution shown, risk buckets created.                 |
| F-05 | Inventory Optimization | EOQ, safety stock, reorder point by SKU.                       | 4,070 products analyzed, top reorder priorities chart shown.                |
| F-06 | Dashboard UI           | Streamlit multi-page interface with charts and exports.        | Sidebar navigation, filters, download buttons, responsive layout.           |

## 3. Technology Stack

| Category    | Technology              | Rationale / Alternatives                        |
|-------------|-------------------------|-------------------------------------------------|
| Language    | Python 3.11+            | Standard for analytics, ML, and Streamlit       |
| Data        | Pandas, NumPy           | Fast tabular processing and numeric computation |
| ML          | Scikit-learn, XGBoost   | Reliable clustering and classification          |
| Forecasting | Prophet, PyTorch / LSTM | Seasonality modelling + neural time-series      |
| Dashboard   | Streamlit, Plotly       | Rapid interactive UI with rich charts           |
| Deployment  | Streamlit Cloud         | Simple web deployment for dashboards            |
| Data I/O    | openpyxl                | Load Excel retail source files                  |

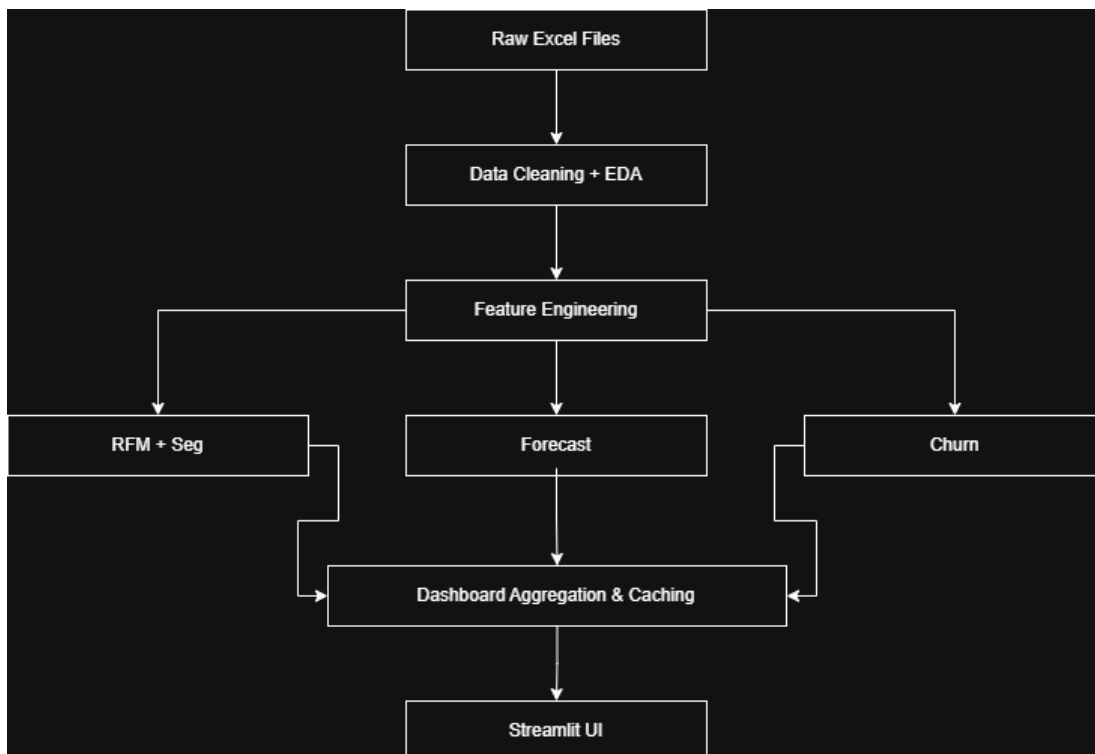
## 4. Architecture Overview

### High-Level Data Flow

1. Source ingest: Online Retail.xlsx + online\_retail\_II.xlsx
2. Data cleaning & feature engineering
3. RFM scoring and customer clustering
4. Demand time-series aggregation and forecasting
5. Churn modelling and risk segmentation
6. Inventory metrics generation (EOQ / Safety Stock)
7. Streamlit dashboard consumption

### MLOps Pipeline

- Notebook-driven ETL / modelling
- Generated CSV outputs in data/
- Dashboard loads processed data with caching
- Deployment managed via lightweight requirements and Streamlit app



## **5. Detailed Execution Timeline**

### **Week 1**

- Data ingestion from both retail files
- Initial EDA and quality checks
- Exported processed dataset processed\_eda.csv

### **Week 2**

- Evaluated K-Means and DBSCAN
- Forecasting and churn models ready
- Optimization logic implemented

### **Week 3**

- Developed demand forecasting pipeline
- Fully interactive dashboard with all insights
- Produced forecast metrics and comparative results

### **Week 4**

- Final QA,
- README polishing
- Demo video recording

### **Milestones & Checkpoints**

- EDA complete
- Feature engineering complete
- Segmentation complete
- Forecasting models complete
- Churn model complete
- Inventory optimization complete

## **6. Technical Highlights**

### **Model Performance**

#### **Customer Segmentation**

- K-Means clusters: 6
- Silhouette Score: 0.5923
- Davies-Bouldin: 0.6461

#### **Forecasting**

- Prophet MAPE: 47.01%
- LSTM MAPE: 99.94%
- Ensemble MAPE: 56.20%
- $R^2$ : Prophet -0.38, Ensemble -1.53

#### **Churn Prediction**

- AUC-ROC: 1.00
- Precision: 1.00
- Recall: 1.00
- F1-score: 1.00

#### **Feature Engineering**

- Combined two retail Excel sources into one dataset
- Created RFM features and scores
- Built lag, rolling window, and temporal features for forecasting
- Normalized features for model training
- Derived issuance of churn labels using recency thresholds

#### **MLOps Practices**

- Modular notebook pipeline for reproducibility
- Generated cached CSV artifacts for dashboard load speed

- Added Streamlit caching for expensive retail Excel loading
- Documented model outputs and metrics in README

## **Challenges & Solutions**

- Large Excel load time: solved with cached data and cached inventory outputs
- Mixed data types in source files: handled with dtype cleaning in feature engineering
- Forecast model instability: retained Prophet baseline and ensemble results for robustness
- Real product inventory usage: replaced synthetic inventory with actual Online Retail SKUs

## **7. Deployment & Operations**

### **Platform Used**

- Streamlit Cloud for web deployment
- Local development on Windows with Python and notebooks

### **CI/CD Summary**

- Use GitHub push to update dashboard source
- Deployable app uses lightweight dependency file
- .gitignore ensures large Excel files and model artifacts are excluded

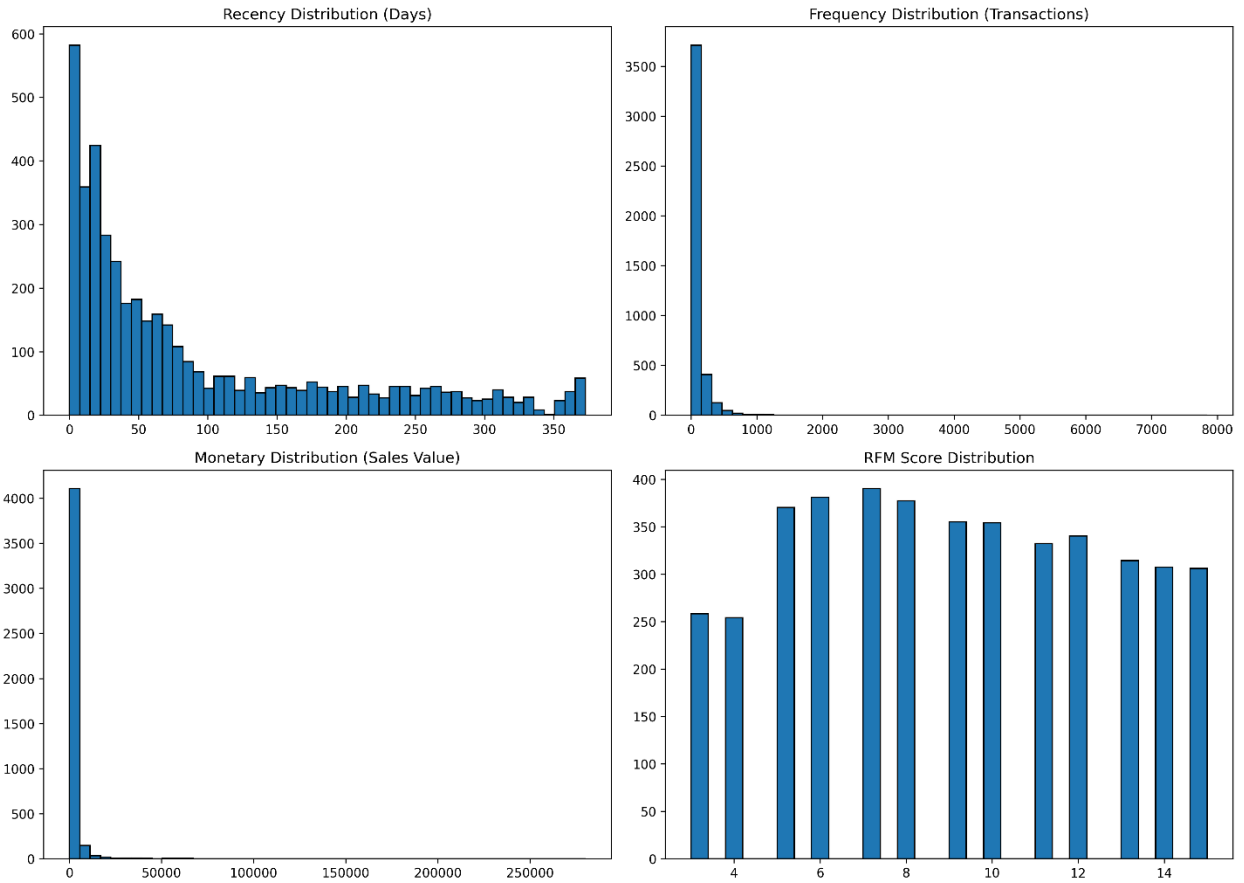
### **Monitoring / Drift Detection**

- Current setup is manual notebook validation
- Dashboard alternatives: add payload logging and drift alerts in future
- Recommended next step: track daily forecast residuals and inventory reorder accuracy

## 8. Visuals

Suggested visuals to include:

- RFM Score distribution



- DBSCAN Clustering

```
=====
DBSCAN CLUSTERING (eps=1.0, min_samples=5)
=====

DBSCAN Results:
- Number of Clusters: 2
- Number of Noise Points: 28
- Cluster Distribution:
DBSCAN_Cluster
-1      28
 0    4310
Name: count, dtype: int64
- Silhouette Score: 0.8948
```



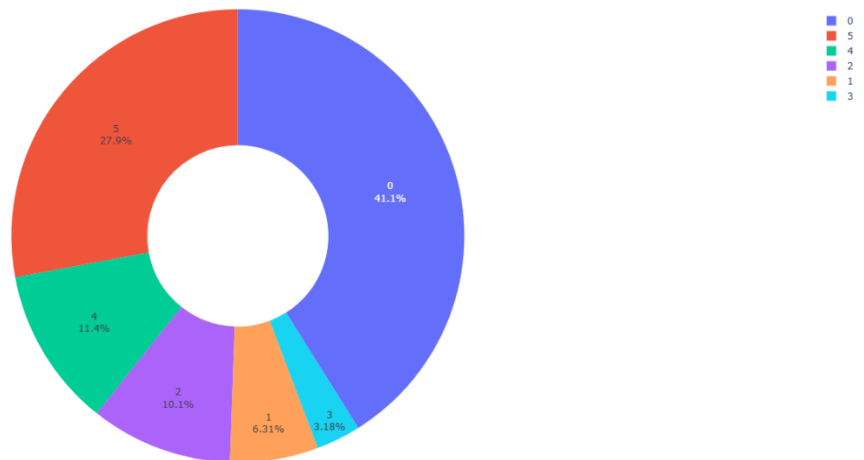
- Kmeans Cluster with k=6

```
K-Means Evaluation Metrics:
- Silhouette Score: 0.5923 (closer to 1 is better)
- Davies-Bouldin Index: 0.6461 (lower is better)
- Calinski-Harabasz Index: 3890.1897 (higher is better)

Cluster Distribution:
KMeans_Cluster
0      2922
1      1056
2         4
3         4
4        14
5       338
Name: count, dtype: int64
```

- Customer segmentation scatter plots

Customer Segment Revenue Share



- Prophet Model

```
Prophet Model Performance:
- MAPE: 0.3801
- MAE: 16008.6897
- RMSE: 20714.8194
- R2: -0.2635
```

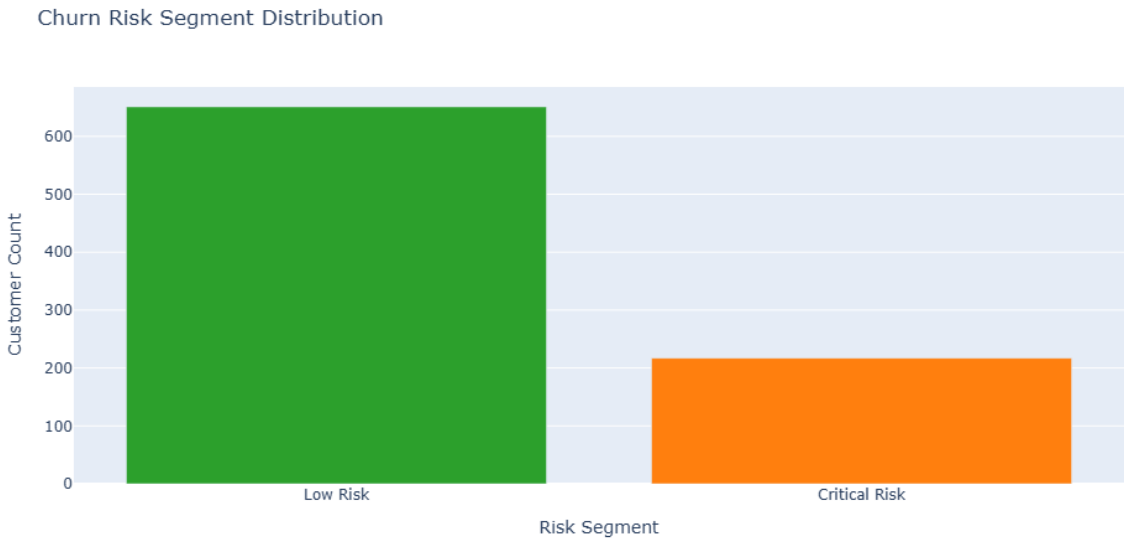
- LSTM Model

```
LSTM Model Performance:  
- MAPE: 0.9994  
- MAE: 57139.3555  
- RMSE: 63616.507  
- R2: -4.1743
```

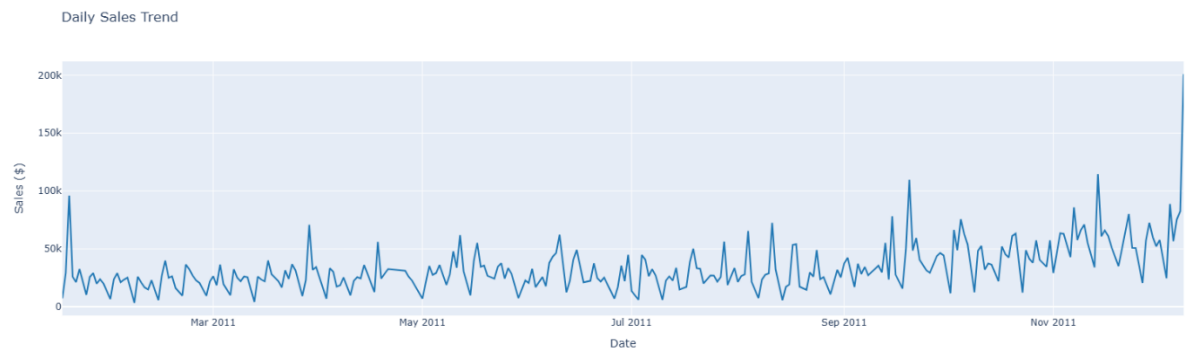
- Churn risk segmentation

```
Churn Risk Segmentation:  
Risk_Segment  
Low Risk      651  
Critical Risk  217  
Medium Risk   0  
High Risk     0  
Name: count, dtype: int64  
  
Risk Segment Characteristics:  
                Recency Frequency Monetary Churn_Probability  
                mean      mean      mean                min  max mean  
Risk_Segment  
Low Risk      40.21    103.58   2145.83                0.0  0.0  0.0  
Medium Risk   NaN      NaN     NaN                 NaN  NaN  NaN  
High Risk     NaN      NaN     NaN                 NaN  NaN  NaN  
Critical Risk 252.69    29.05    859.64                1.0  1.0  1.0
```

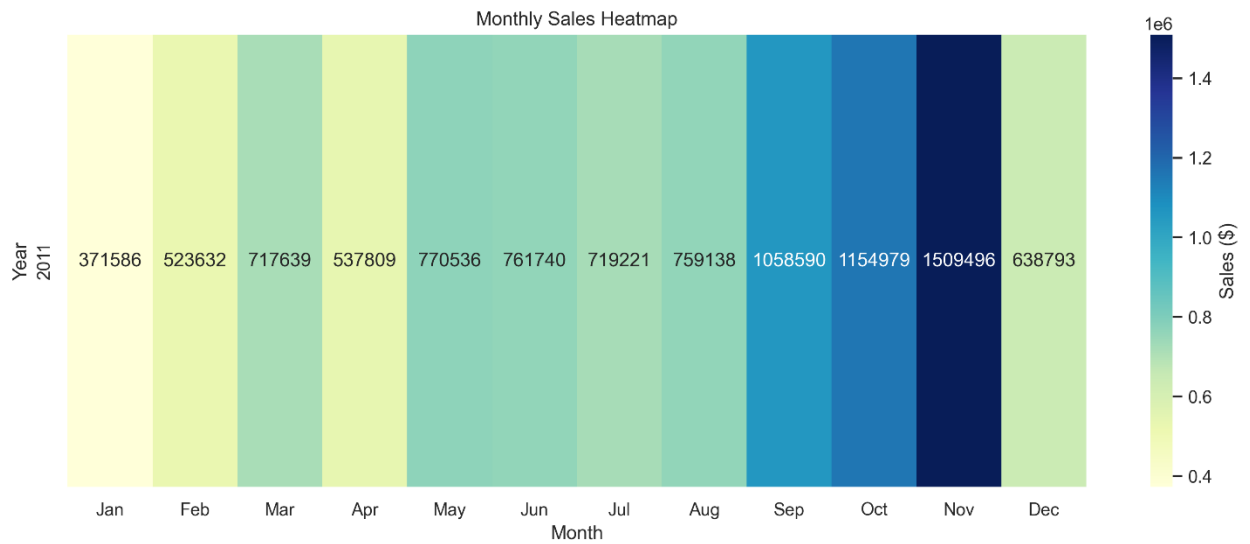
- Churn risk bar chart



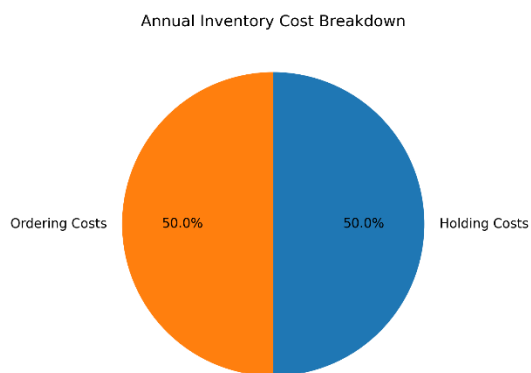
- Daily sales Trend



- Monthly sales heatmap

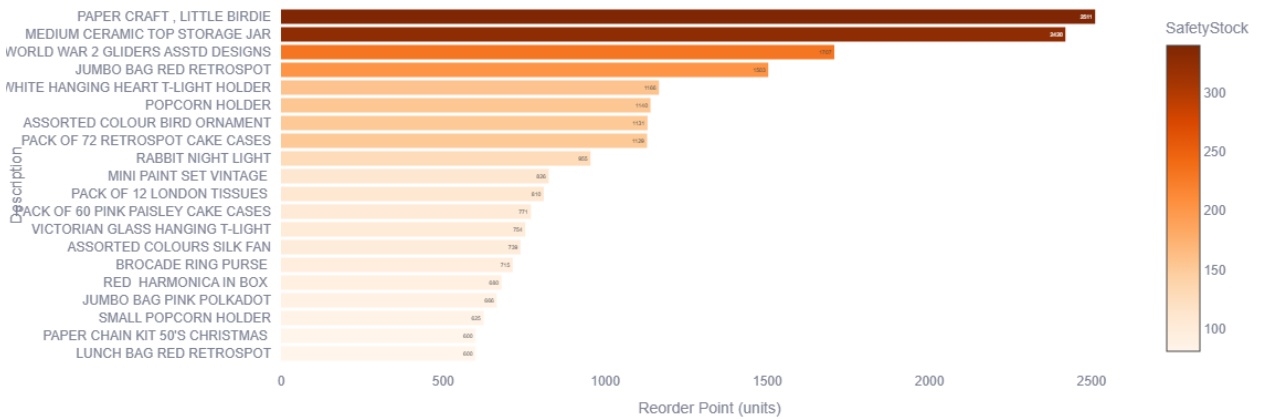


- Inventory reorder priority bar chart



- Reorder Priority

Reorder Priorities by Product



## 9. Personal Reflection

### Key Learnings

- Real retail data requires careful cleaning and merged-source handling.
- RFM segmentation adds strong business meaning to customer clusters.
- Forecast metrics must be interpreted in context: sales dollars vs. units.
- Streamlit caching is essential for usability when loading large Excel files.

### Industry Best Practices Applied

- Use of repeatable notebooks for each pipeline stage
- Separate storage of processed CSV artifacts
- Interactive dashboard with export and filter support
- Clear business-facing segment descriptions

### Future Roadmap Ideas

- Add automated drift detection and retraining triggers, Build a proper CI/CD pipeline for model deployment
- Extend forecasting with advanced models like ARIMA/Prophet hybrid, Add live inventory alerts and reorder notifications